Orchestrating federated processing within a federation of cancer imaging and clinical data nodes

The EUCAIM perspective

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EUCAIM

The EUropean Federation for CAncer IMages (EUCAIM) project unites **76 partners** to create a Europe-wide digital infrastructure of FAIR de-identified cancer images. EUCAIM will preserve data sovereignty and offer a platform, including an "Atlas of Cancer Images", for developing and benchmarking AI tools for Precision Medicine.

AI4HI

Converging into EUCAIM's digital infrastructure, the five AI4HI projects reunite the largest pan-European Atlas of Cancer Images.

PRIMAGE – Medical imaging Artificial intelligence Childhood cancer research
EUCAN IMAGE
ProCAncer-I
INCISIVE
chAIMeleon

The Federation

The project has started with an initial set of **23 clinical sites**, including 19 major university hospitals, from 12 European countries. It aims to expand to at least 30 distributed data providers from 15 countries by 2026.

Central Infrastructure

Dashboard. For researchers and providers to access and manage the datasets, with guidance and tools for datasets preparation.

AAI. Manages user access and authorization based on roles and identity verification, controlling access to metadata and platform resources (including federated processing).

Public catalogue. Stores metadata for datasets discovery, showing datasets descriptions and access conditions.

Federated query. Sends and aggregates queries across the federation's infrastructure, providing datasets status and access.

Resource negotiation. Tools for submitting, refining, and approving access requests, integrating "Data Access Committee"'s decisions for role-based access control.

Access to data. A) *Full data download*, B) *Full on-premises access*, and C) *Restricted on-premises access*. Each with specific conditions for viewing and processing data.

Federated processing. Enables collaborative data processing without direct data sharing, using decentralized architecture.

Central repository. Stores data that cannot be kept locally, including imaging and non-imaging data, for collaborative research and access by external parties.

What is federated processing?

Federated (data) processing describes a specific analytical methodology performed on multiple separated datasets. The typical characteristic of federated processing is its decentralized data infrastructure. During this analysis, the data is not exchanged and stays behind the owner's technical and legal barriers. Only analytical parameters are exchanged between the data-hosting sites, and these parameters are not allowed to contain information that may endanger the datasets.

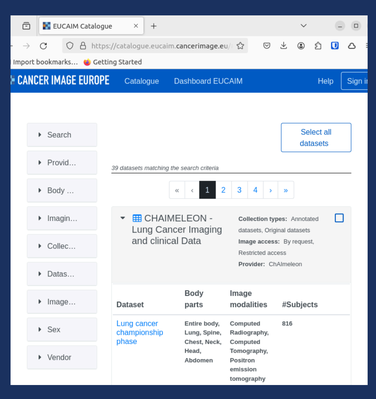
What is federated learning?

Federated learning, a specific case of federated (data) processing, is a field of machine learning focusing on training models on a decentralized data infrastructure by multiple entities collaboratively. Taking advantage of the federated processing infrastructure for the model training process, federated learning ensures that the data used in model training remains local and private, in contrast to machine learning settings in which data is centrally stored.

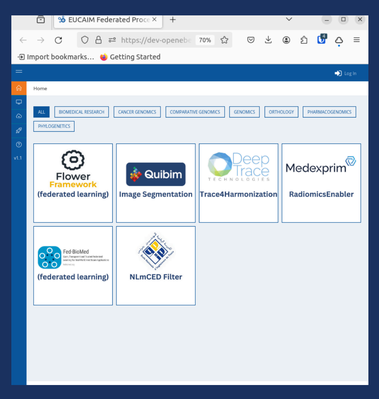
Federated processing and federated learning in EUCAIM

01 Select dataset(s) of interest

At the data catalogue, the user applies a series of filters to narrow down the datasets of interest



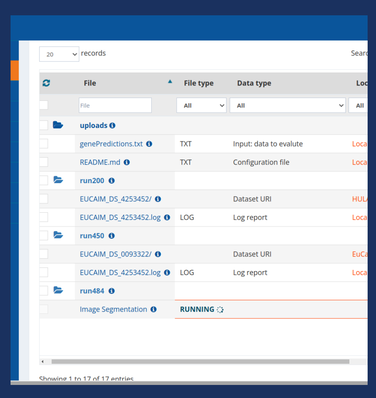
02 Select and configure the tool



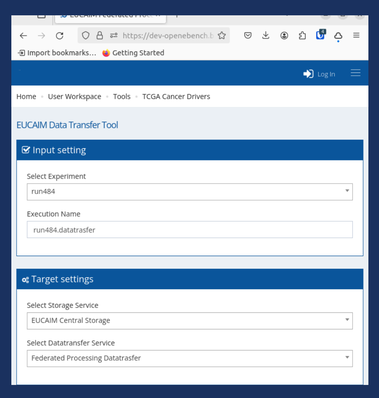
From the data catalogue, the user selected the Federated Processing options, accessing the dashboard. At the dashboard, the user can select and configure the tool to apply to the datasets of interest.

03 Run the experiment

Once the "experiment" (combination of dataset(s) and tool(s)) is set. The user can review it, identify the status of the sites having the data, and run the experiment.



04 Export or store the results



Once the experiment is finished. The user can check the ending status and the location of the output through the dashboard, allowing the results to be exported or stored at the EUCAIM central storage.

Components of the EUCAIM's Federated Processing architecture

Analysis orchestrator

It is composed by the Federated Processing dashboard, a web page to properly set, configure and check the status of the experiments, the analysis orchestrator, a python engine deploy at the EUCAIM central infrastructure in charge to manage the multiple experiments run at the platform, and the analysis deamon, a deamon located at each member of the federation in charge to manage the experiments run in that specific node.

Data materializer

It is a user-transparent tool, run by the analysis orchestrator at the beginning of each experiment, that interacts with the data materialization service specific to each node to ensure the data is accessible by the user and it is available for the experiment.

Data transference

It is a user-transparent tool with two functionalities: a) for those nodes with an agreement with 3rd party infrastructures, it copies the data from the node to the computation infrastructure; b) once any experiment finishes, it moves the results from the computational infrastructure - local to the node of from a 3th party - to EUCAIM's central storage.

Analytical framework

Agnostic layer for analytical purposes, EUCAIM is betting for two frameworks for federated learning: the American Flower and the European Fed-BioMed. Any analysis that can be encapsulated in a docker-like definition can be run using the both frameworks, and multiple experiment using PyTorch, TensorFlow, Scikit-learn have been run over any f the two.

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Support

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